

SCQ (Single Correct Type) :

- The number $N = {}^{20}C_7 - {}^{20}C_8 + {}^{20}C_9 - {}^{20}C_{10} + \dots - {}^{20}C_{20}$ is not divisible by :
 (A) 3 (B) 7 (C) 11 (D) 19
- In the expansion of $(1+x)^2(1+y)^3(1+z)^4(1+w)^5$, the sum of the coefficient of the terms of degree 12 is:
 (A) 61 (B) 71 (C) 81 (D) 91
- If $\sum_{r=0}^n \left(\frac{r^3 + 2r^2 + 3r + 2}{(r+1)^2} \right) {}^nC_r = \frac{2^4 + 2^3 + 2^2 - 2}{3}$ then the value of n is :
 (A) 2 (B) 2^2 (C) 2^3 (D) 2^4
- The ratio in which the area bounded by curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$ is :
 (A) 7 : 15 (B) 15 : 49 (C) 1 : 3 (D) 17 : 49
- For $0 < r < 1$, let n_r denotes the line that is normal to the curve $y = x^r$ at the point $(1, 1)$. Let S_r denotes the region in the first quadrant bounded by the curve $y = x^r$; the x-axis and the line n_r . Then the value of r that minimizes the area of S_r is :
 (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2} - 1$ (C) $\frac{\sqrt{2}-1}{2}$ (D) $\sqrt{2} - \frac{1}{2}$
- The area bounded by $|x| = 1 - y^2$ and $|x| + |y| = 1$ is :
 (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) 1
- Let $f(x) = x^3 - 3x^2 + 3x + 1$ and g be the inverse of it ; then area bounded by the curve $y = g(x)$ with x-axis between $x = 1$ to $x = 2$ is (in square units) :
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) 1
- The area of the loop formed by $y^2 = x(1-x^3)$ is
 (A) $\int_0^1 \sqrt{x-x^4} dx$ (B) $2 \int_0^1 \sqrt{x-x^4} dx$
 (C) $\int_{-1}^1 \sqrt{x-x^4} dx$ (D) $4 \int_0^{1/2} \sqrt{x-x^4} dx$
- The area enclosed between the curves $|x| + |y| \geq 2$ and $y^2 = 4\left(1 - \frac{x^2}{9}\right)$ is :
 (A) $(6\pi - 4)$ sq. units (B) $(6\pi - 8)$ sq. units (C) $(3\pi - 4)$ sq. units (D) $(3\pi - 2)$ sq. units

10. Area enclosed by the curves $y = x^2 + 1$ and a normal drawn to it with gradient -1 ; is equal to :

- (A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

MCQ (One or more than one correct) :

11. Let $(1 + x^2)^2 (1 + x)^n = \sum_{k=0}^{n+4} a_k x^k$. If $n \in \mathbb{N}$ and a_1, a_2, a_3 are in arithmetic progression then the possible

value(s) of n is/are

- (A) 2 (B) 3 (C) 4 (D) 5

12. If the co-efficient of x^{2r} is greater than half of the co-efficient of x^{2r+1} in the expansion of $(1 + x)^{15}$; then the possible value of ' r ' equal to :

- (A) 5 (B) 6 (C) 7 (D) 8

13. Let $f(x) = 1 + x^{111} + x^{222} + x^{333} + \dots + x^{999}$ then $f(x)$ is divisible by

- (A) $x + 1$ (B) x
(C) $x - 1$ (D) $1 + x^{222} + x^{444} + x^{666} + x^{888}$

14. Consider the equation $\int_0^{c/2} (\cos x - \cos(x-c)) dx = \int_{\pi}^{\frac{\pi}{2}+c} \cos(x-c) dx$ which of the following is subset of solution set of ' c ' for this equation(where ' n ' is an integer)

- (A) $4n\pi + \frac{\pi}{3}$ (B) $4n\pi + \frac{5\pi}{3}$ (C) $2n\pi + \frac{\pi}{6}$ (D) $2n\pi - \frac{\pi}{6}$

15. Suppose $f(x)$ is a real valued differentiable function defined on $[1, \infty)$ with $f(1) = 1$. Further let $f(x)$ satisfy $f'(x) = \frac{1}{x^2 + f^2(x)}$, then the values $f(x)$ are contained in interval

- (A) $[1, \infty)$ (B) $[1, 1 + \pi/4)$ (C) $(\pi/4, 1]$ (D) $[1 - \pi/4, 1]$

16. The value of $\int_0^{\infty} 4[x+7]e^{-2x} dx$ is $\frac{ae^2 - 12}{e^2 - b}$ then (Here $[.]$ represents greatest integer function).

- (A) $a = 14$ (B) $a = 13$ (C) $b = 2$ (D) $b = 1$

17. If $I_1 = \int_0^1 \frac{1+x^8}{1+x^4} dx$ and $I_2 = \int_0^1 \frac{1+x^9}{1+x^3} dx$, then

- (A) $I_2 < I_1 < \pi/4$ (B) $\pi/4 < I_2 < I_1$ (C) $1 < I_1 < I_2$ (D) $I_2 < I_1 < 1$

18. If $g(x) = \{x\}^{[x]}$, where $\{.\}$ and $[.]$ represents fractional part and greatest integer function respectively and

$$f(k) = \int_k^{k+1} g(x) dx \quad (k \in \mathbb{N}), \text{ then}$$

(A) $f(1), f(2), f(3), \dots$ are in H.P.

$$(B) \sum_{r=1}^{\infty} (-1)^{r+1} f(r) = 1 - \ell n 2$$

$$(C) \sum_{r=1}^{\infty} (-1)^r f(r) = \ell n \left(\frac{2}{e} \right)$$

$$(D) \sum_{r=0}^{\infty} f\left(\frac{1}{r}\right) = \frac{n(n+1)}{2}$$

19. If $\int_0^1 \frac{x^4 (1+x^{10065})}{(1+x^5)^{2015}} dx = \frac{1}{p}$, then

- (A) Number of ways in which p can be expressed as a product of two relatively prime factors is 8.
 (B) Number of ways in which p can be expressed as a product of two relatively prime factors is 4.
 (C) Number of ways in which p can be expressed as a product of two factors is 8.
 (D) Number of ways in which p can be expressed as a product of two factors is 4.

20. If a curve $y = a\sqrt{x} + bx$ passes through point $(1, 2)$ and the area bounded by curve, line $x = 4$ and x -axis is 8, then :

(A) $a = 3$

(B) $b = 3$

(C) $a = -1$

(D) $b = -1$

Numerical Based Questions :

21. Let the sum of all divisors of the form $2^p \cdot 3^q$ (with p, q positive integers) of the number $19^{88} - 1$ be λ . Find the unit digit of λ .

22. Let e be the eccentricity of hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola then find

the value of $\left\{ \int_1^{\sqrt{2}} (f(e) + f(f(e))) de \right\}$

23. Let f be a monic polynomial function of degree n ($n \geq 1$) such that $\int_0^{\pi/2} \sin^4 x dx < \int_0^{\pi/2} (\sin x)^n dx < 1$. If $f(3) =$

$f''(3) = f(4) - 4 = 0$, then find the value of $\frac{1}{2} \left(\int_3^5 f(x) dx \right)$.

24. Let $I_n = \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n} \right) dx$ where $n \in \mathbb{N}$. If $\lim_{n \rightarrow \infty} I_n$ can be expressed as a rational

number $\lim_{n \rightarrow \infty} I_n$ in the lowest form, then find the value of $p + q$.



ANSWERKEY OF CAPS – 8+

1.	(C)	2.	(D)	3.	(A)	4.	(B)	5.	(B)	6.	(C)	7.	(B)
8.	(B)	9.	(B)	10.	(D)	11.	(ABC)	12.	(ABC)	13.	(AD)	14.	(AB)
15.	(AB)	16.	(AD)	17.	(BD)	18.	(ABC)	19.	(AC)	20.	(AD)	21.	4
22.	3	23.	5	24.	5								